

Nathan Critchett

COGNITIVE STRATEGIST · AI BUILDER · AUTHOR

I work at the seam between what AI can produce and what the humans using it can actually evaluate — designing the systems, products, and training that make people sharper with AI instead of dependent on it.

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SUMMARY

Part strategist, part builder. I turn cognitive-science research into **shipped products and real adoption** — two live platforms, an interactive diagnostic engine, and a serverless content/automation stack, all built solo. The thinking behind them comes from field work inside **100+ organizations**, distilled into a forthcoming book and a library of **60+ evidence-based articles**. I move comfortably between writing the strategy, prototyping the tool, designing the learning, and standing on stage to teach it.

SELECTED AI TOOLS, PRODUCTS & PROTOTYPES

Arq.Training — Founder & Builder

● Live · [arq.training](#)

A cognitive diagnostic platform for education. Curriculum and test scores go in; a visible map of *how students actually reason* comes out — the thinking in between that standardized tests miss. Built on the Model of Hierarchical Complexity and adult-development research.

CritchPitch — Founder & Builder

● Live · [critchpitch.com](#)

Puts a pitching-biomechanics lab in a parent's pocket: screens a young pitcher's mechanics and arm-injury risk from a single phone video. Paired with a 25-article, fully cited library on youth arm health.

The Cognitive Audit — Designer & Developer

● Live · [/audit](#)

An interactive 15-question diagnostic web app that scores an organization against five frameworks, weights the responses, surfaces the weakest link, and returns a single prioritized recommendation.

Custom scoring engine · results UI · email-gated reporting

AI-Native Portfolio CRM — Prototype

In development

A context-extraction system that mines expert context across an entire portfolio of companies and surfaces the patterns that matter — keeping a human at the center of every conversation.

Architects List — Serverless backend

Shipped

Email-capture and automated welcome-kit delivery powering the book and audit funnels.

Cloudflare Workers · KV storage · Resend email API · auth-gated dashboard

A hand-built publishing system — designed, written, and engineered solo — with a programmatic build pipeline that generates Open Graph cards, sitemaps, and schema.org structured data across three libraries.

Python + Node build tooling · structured data · analytics

AUTHOR & PUBLISHED RESEARCH

Cognitive Architecture: How to Think When Machines Think For You — Author (2026). A short field guide for leaders, built from data inside 100+ organizations. Five frameworks: the Cognitive Supply Chain, the Centaur Architecture, the Antifragile Ego, the Horizontal/Vertical distinction, and the identity shift from worker to Architect of Intelligence.

8 whitepapers + 16 essays on AI & cognition — including *The Complexity Gap*, *Cognitive Offloading*, *Universal Basic Upgrading*, *Stop Grading AI Output*, and *Two Careers AI Can't Kill*. Grounded in cognitive science (Kegan, Commons, cognitive-load theory) and tested in the field.

60+ evidence-based articles across three libraries — AI & Cognition (24), Cognitive Development / Arq (14), and Youth Pitching / CritchPitch (25), each fully sourced.

SPEAKING & ADVISORY

Keynotes, half- and full-day sessions, and professional development on cognitive strategy for the AI era. Signature talks: **The Complexity Gap**, **AI as Mirror, Not GPS**, and **Building the Thinkers the Future Needs**. District work delivered in partnership with **Edapt** and Christian Jackson. Selected audiences:

- ◆ **Greater Ozarks Cooperating School Districts** (GOCSO)
- ◆ **AAEA** — educational administrators
- ◆ **LA County Fire Department** — Lifeguard Division
- ◆ **Association of California School Administrators** (ACSA)
- ◆ **Small School Districts' Association** (SSDA)
- ◆ **Multiple K-12 school districts**

VENTURES FOUNDED

Arq.Training — Founder. Cognitive diagnostics for education.

CritchPitch — Founder. Youth pitching screening & arm-health.

CORE SKILLS

- AI strategy & adoption
- Learning & cognitive-development design
- Rapid prototyping
- LLM-assisted product design
- Serverless (Cloudflare Workers)
- JavaScript
- Python build tooling
- Diagnostic & assessment design
- Programmatic SEO & content systems
- Keynote speaking & PD facilitation
- District & stakeholder engagement
- Technical writing

EARLIER

Division I pitcher & sports-medicine All-American. Carried the same intensity into a new problem: building the thinkers the future actually needs. **Former first responder** — LA County Fire Department, Lifeguard Division (ocean rescue).